Customer Churn Prediction using Machine Learning and Explainable AI (SHAP)

Project Overview

This project develops an end-to-end machine learning solution to predict customer churn for a fictional telecommunications company using demographic, account, billing and service usage data. The workflow covers data preparation, exploratory analysis, model development, hyperparameter tuning, explainable AI and business recommendations.

Business Problem

Customer churn reduces recurring revenue and increases customer acquisition costs. The objective is to identify customers at risk of leaving so the business can proactively retain them.

Objectives

Predict customer churn, understand key churn drivers, compare Logistic Regression, Random Forest and XGBoost, optimize models using GridSearchCV, interpret predictions with SHAP, and provide actionable business recommendations.

Dataset Overview

Dataset contains 7,043 customer records with 21 variables including demographics, account information, subscribed services and the target variable (Churn).

Methodology

Project stages: data loading, cleaning, EDA, feature engineering, encoding, scaling, train-test split, baseline models, hyperparameter tuning, final model selection and SHAP explainability.

Data Preparation

Converted column names to snake_case, converted total_charges to numeric, handled blank values, encoded categorical variables using one-hot encoding, scaled numerical variables for Logistic Regression and created train/test datasets.

Exploratory Data Analysis

EDA explored churn distribution, tenure, contract type, monthly charges, payment methods, internet services and correlation patterns to identify important business trends.

Machine Learning Models

Three supervised classification algorithms were developed: Logistic Regression (baseline), Random Forest (ensemble learning) and XGBoost (gradient boosting). Performance was evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC.

Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation optimized Logistic Regression, Random Forest and XGBoost using ROC-AUC as the scoring metric.

Explainable AI

SHAP Summary, Beeswarm, Bar, Waterfall, Force and Dependence plots were used to explain global and local feature importance and improve model transparency.

Business Insights

Customers with month-to-month contracts, shorter tenure, higher monthly charges and limited value-added services are generally at higher risk of churn, while longer contracts and additional support services improve retention.

Recommendations

Promote long-term contracts, strengthen onboarding for new customers, offer personalized pricing and bundles, encourage adoption of support/security services and target high-risk customers with retention campaigns.

Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, SHAP and Jupyter Notebook.

Conclusion

The project demonstrates a complete machine learning lifecycle from raw data to explainable predictions. It combines predictive analytics with business interpretation to support customer retention strategies and data-driven decision making.